

Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

The Data Flow Diagram (DFD) below depicts the flow of information within the India's Agriculture Crop Production Analysis system - showing how raw agricultural data (rainfall, area, season, soil, and historical yield records) enters the system, how it is processed and analyzed using the machine learning model, and how the predicted crop production / yield results are delivered to the end users.

DFD Level 0 (Context Level):

Farmer / User → provides input (State, District, Season, Crop, Area, Rainfall) → **Crop Production Analysis System** → fetches historical data from **Dataset / Database** → **ML Prediction Engine** processes the request → predicted production / yield → returned to Farmer / User via the **Web Application UI**. The **Administrator** manages the dataset, retrains the model, and monitors system usage.

DFD Level 1 (Detailed):

1. User registers / logs into the web application.
2. User enters crop details - State, District, Season, Crop type, Area, Year and Rainfall.
3. Application validates the input and sends the request to the prediction service.
4. Prediction service loads the trained ML model (Random Forest / Regression) and the historical crop dataset from the database.
5. Model computes the predicted crop production / yield.
6. Result is sent back to the application and displayed to the user as a report / chart (production in tonnes, yield per hectare).
7. Administrator can upload new datasets, trigger model retraining, and view analytics dashboards (state-wise / crop-wise production trends).

User Stories:

The following user stories have been identified for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Farmer / End User (Web user)	Registration	USN-1	As a user, I can register for the application by entering my name, email, password, and confirming my password.	I can register and access my dashboard	High	Sprint-1
		USN-2	As a user, I will receive a confirmation email once I have registered for the application.	I receive a confirmation email and can click confirm	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
		USN-3	As a user, I can register for the application through Gmail.	I can register and access the dashboard using Gmail login	Medium	Sprint-1
	Login	USN-4	As a user, I can log into the application by entering my email and password.	I can log in and access the dashboard	High	Sprint-1
		USN-5	As a user, I can log into the application through Gmail.	I can log in with Gmail credentials	Low	Sprint-2
	Crop Data Input	USN-6	As a user, I can enter the State, District, Season, Crop and Area to get the predicted crop production.	I can submit crop details through a form	High	Sprint-1
	Prediction / Dashboard	USN-7	As a user, I can view the predicted crop production / yield along with a visual chart on the dashboard.	I can view predicted output and a graph on my dashboard	High	Sprint-2
		USN-8	As a user, I can compare the predicted production of a crop across different years / seasons.	I can view comparison charts for the selected crop	Medium	Sprint-2
Customer Care Executive	Query Handling	USN-9	As a customer care executive, I can view and respond to user queries raised regarding predictions.	I can access and resolve raised tickets	Medium	Sprint-3
Administrator	Dataset Management	USN-10	As an administrator, I can upload / update the historical crop production dataset used for training.	I can upload a dataset file and see a success confirmation	High	Sprint-2
	Model Management	USN-11	As an administrator, I can retrain the ML model with the updated dataset.	I can trigger retraining and view updated model accuracy	High	Sprint-3
	Analytics	USN-12	As an administrator, I can view overall system analytics such as most searched crops and state-wise usage.	I can view an analytics dashboard	Low	Sprint-3